GRADE 12 CHEMISTRY

UNIT 4 — POLYMERS

Monomers, polymerization reactions and classification of polymers

Exam-focused revision booklet

Prepared from the uploaded Ethiopian Grade 12 Chemistry Student Textbook (Ministry of Education, 2023). The explanations below are newly written study notes, not copied textbook pages. Diagrams are original schematic summaries.

How to use this booklet

Read the core idea, memorize the key relationship or equation, redraw the visual summary from memory, then answer both alternative questions without looking. For calculations, show units and working. For process questions, give the stages in the correct order.

CORE CONCEPTS	KEY EQUATIONS / PROCESSES	EXAM PRACTICE

Coverage: 19 focused topic sheets plus this guide page = 20 pages minimum. The two questions on each topic are alternatives for self-testing.

4.01 Meaning of polymers

Core concept

A polymer is a large molecule built from many repeating smaller units. The word polymer means many parts. Polymer molecules may be linear, branched or cross-linked. The number of repeating units is the degree of polymerization; polymer molar mass depends on repeat-unit mass and this number. Examples in the textbook include polyethylene, cellulose and proteins.

Remember

Polymer = many repeating units; DP = number of repeat units

Meaning of polymers	KEY RELATIONSHIP	EXAM APPLICATION

Exam focus

Be ready to define the term precisely, interpret the chemical representation, and explain the reason behind the result or process. Where a calculation is involved, identify the known quantities, choose the correct relationship, substitute with units and check whether the answer is chemically reasonable.

Alternative practice questions

A. Define polymer and degree of polymerization.

B. Classify a polymer chain as linear, branched or network from a structural description.

4.01 • Exam diagram & MCQ practice

Key exam idea: A polymer is a macromolecule made of many repeating units; the number of repeat units is the degree of polymerization.

Diagram reading: Redraw the sequence and explain the role of each labelled part. This is an original revision schematic, not a scale drawing.

Question 1. A polymer is best described as:

A. a large molecule made from repeating units
B. a single atom with a high charge
C. a mixture of two salts
D. a molecule that contains only one carbon

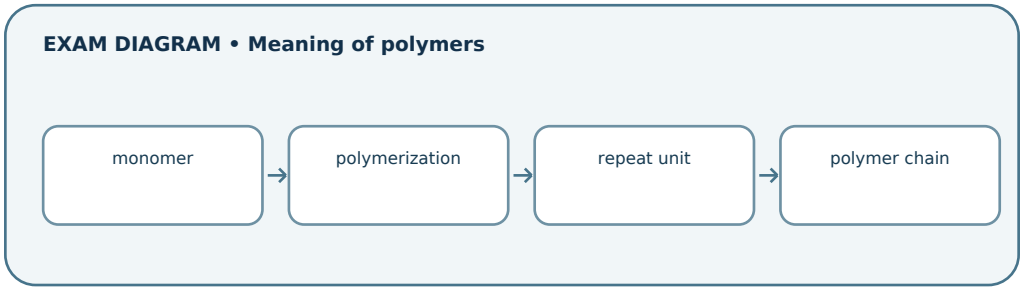
Answer: A. Polymers contain many covalently linked repeating structural units.

Question 2. In the diagram, which labelled step comes immediately after “monomer”?

A. polymerization
B. repeat unit
C. polymer chain
D. monomer

Answer: A. The schematic is arranged in sequence: monomer → polymerization → repeat unit → polymer chain.

Exam reminder: Learn the chemical meaning of the arrows and labels; do not memorize a flowchart without understanding the reaction or process.



4.02 Monomers and repeat units

Core concept

A monomer is a small molecule that acts as a building block for a polymer. During polymerization, monomer units become linked. The repeat unit is the structural pattern repeated along the polymer chain and often corresponds to the monomer after bond changes. Examples include ethene for polyethylene, glucose units for cellulose and amino acids for proteins.

Remember

Monomer → repeating units → polymer

Monomers and repeat units	KEY RELATIONSHIP	EXAM APPLICATION

Exam focus

Be ready to define the term precisely, interpret the chemical representation, and explain the reason behind the result or process. Where a calculation is involved, identify the known quantities, choose the correct relationship, substitute with units and check whether the answer is chemically reasonable.

Alternative practice questions

A. Identify the monomer used to make polyethylene.

B. Give examples of monomers associated with cellulose and proteins.

4.02 • Exam diagram & MCQ practice

Key exam idea: A monomer is a small molecule that can join with others; a repeat unit is the recurring structural segment in the polymer.

Diagram reading: Redraw the sequence and explain the role of each labelled part. This is an original revision schematic, not a scale drawing.

Question 1. The monomer used to make polyethylene is:

A. ethene
B. ethanol
C. ethanoic acid
D. ethyne

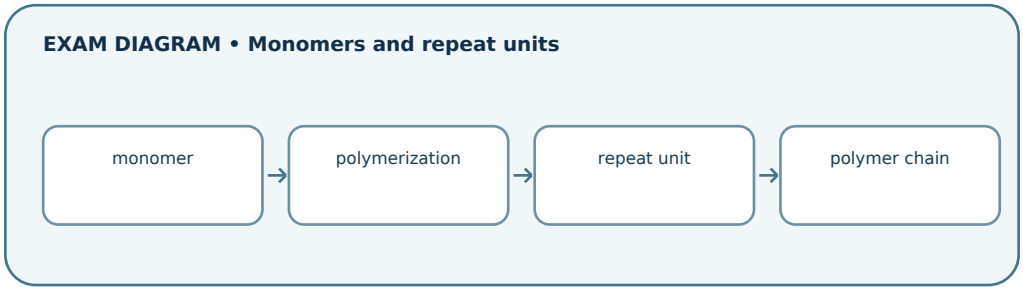
Answer: A. Ethene's C=C bond opens during addition polymerization to form the polyethylene chain.

Question 2. In the diagram, which labelled step comes immediately after “monomer”?

A. polymerization
B. repeat unit
C. polymer chain
D. monomer

Answer: A. The schematic is arranged in sequence: monomer → polymerization → repeat unit → polymer chain.

Exam reminder: Learn the chemical meaning of the arrows and labels; do not memorize a flowchart without understanding the reaction or process.



4.03 Polymer architecture

Core concept

Polymer chains may be linear, slightly branched or highly interconnected. Bifunctional monomers can form linear chains, while monomers with functionality greater than two can create cross-linked networks. Architecture influences physical behavior and response to heating. Use structural descriptions to explain differences rather than relying only on names.

Remember

Linear / branched / cross-linked network

Polymer architecture	KEY RELATIONSHIP	EXAM APPLICATION

Exam focus

Be ready to define the term precisely, interpret the chemical representation, and explain the reason behind the result or process. Where a calculation is involved, identify the known quantities, choose the correct relationship, substitute with units and check whether the answer is chemically reasonable.

Alternative practice questions

A. Explain how monomer functionality affects polymer structure.

B. Compare a linear polymer with a cross-linked polymer.

4.03 • Exam diagram & MCQ practice

Key exam idea: Polymer chains may be linear, branched or cross-linked; cross-linking connects chains into a network.

Diagram reading: Redraw the sequence and explain the role of each labelled part. This is an original revision schematic, not a scale drawing.

Question 1. Which structure contains links connecting separate polymer chains?

A. cross-linked network
B. isolated monomers
C. only a single unconnected chain
D. a simple ionic lattice

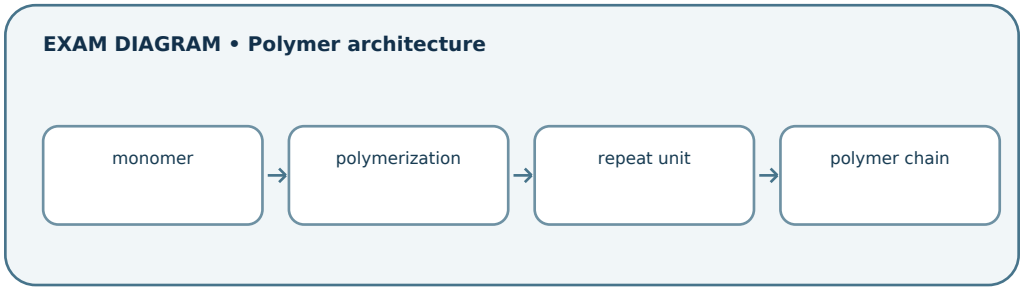
Answer: A. Cross-links join polymer chains and produce a network structure.

Question 2. In the diagram, which labelled step comes immediately after “monomer”?

A. polymerization
B. repeat unit
C. polymer chain
D. monomer

Answer: A. The schematic is arranged in sequence: monomer → polymerization → repeat unit → polymer chain.

Exam reminder: Learn the chemical meaning of the arrows and labels; do not memorize a flowchart without understanding the reaction or process.



4.04 Polymerization overview

Core concept

Polymerization is the chemical process by which monomers combine to form polymers. The textbook emphasizes addition and condensation polymerization. To classify a reaction, inspect the monomer functional groups and whether a small molecule is eliminated. Draw the monomer and repeating unit, then show how the chain grows.

Remember

Monomer units combine → polymer

Polymerization overview	KEY RELATIONSHIP	EXAM APPLICATION

Exam focus

Be ready to define the term precisely, interpret the chemical representation, and explain the reason behind the result or process. Where a calculation is involved, identify the known quantities, choose the correct relationship, substitute with units and check whether the answer is chemically reasonable.

Alternative practice questions

A. Define polymerization and name its two main types.

B. What structural feature of a monomer can allow it to link to more than one molecule?

4.04 • Exam diagram & MCQ practice

Key exam idea: Polymerization joins monomers to form polymers; the two broad classes are addition and condensation polymerization.

Diagram reading: Redraw the sequence and explain the role of each labelled part. This is an original revision schematic, not a scale drawing.

Question 1. Which pair names the two main polymerization types?

A. addition and condensation
B. oxidation and reduction
C. neutralization and hydrolysis
D. evaporation and sublimation

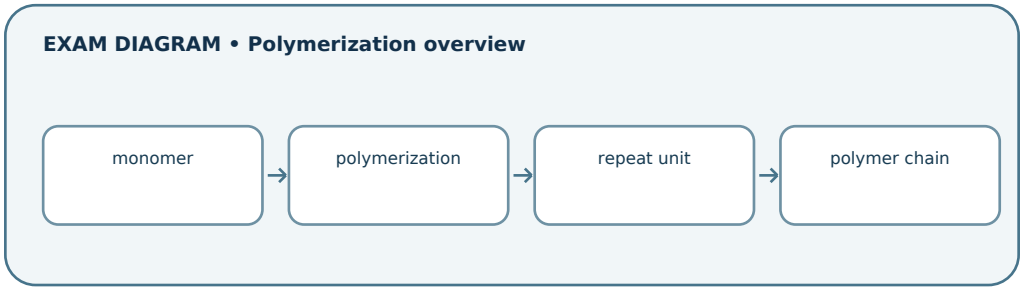
Answer: A. Addition and condensation describe the two broad reaction pathways covered in the unit.

Question 2. In the diagram, which labelled step comes immediately after “monomer”?

A. polymerization
B. repeat unit
C. polymer chain
D. monomer

Answer: A. The schematic is arranged in sequence: monomer → polymerization → repeat unit → polymer chain.

Exam reminder: Learn the chemical meaning of the arrows and labels; do not memorize a flowchart without understanding the reaction or process.



4.05 Addition polymerization

Core concept

Addition polymerization is chain-growth polymerization in which unsaturated monomers add to a growing chain. A carbon–carbon double bond is converted into single bonds as the chain forms. The molecular formula of the repeat unit corresponds to the monomer composition; no small molecule is eliminated in the basic addition process.

Remember

n CH■=CH■ → [–CH■–CH■–]■

Addition polymerization	KEY RELATIONSHIP	EXAM APPLICATION

Exam focus

Be ready to define the term precisely, interpret the chemical representation, and explain the reason behind the result or process. Where a calculation is involved, identify the known quantities, choose the correct relationship, substitute with units and check whether the answer is chemically reasonable.

Alternative practice questions

A. Define addition polymerization.

B. Explain what happens to the C=C bond during ethene polymerization.

4.05 • Exam diagram & MCQ practice

Key exam idea: Addition polymerization joins unsaturated monomers without eliminating a small molecule; the C=C bond becomes part of the backbone.

Diagram reading: Redraw the sequence and explain the role of each labelled part. This is an original revision schematic, not a scale drawing.

Question 1. During addition polymerization of ethene, the C=C bond:

A. opens to form new C–C single bonds
B. remains unchanged in every repeat unit
C. forms an ester linkage
D. releases carbon dioxide

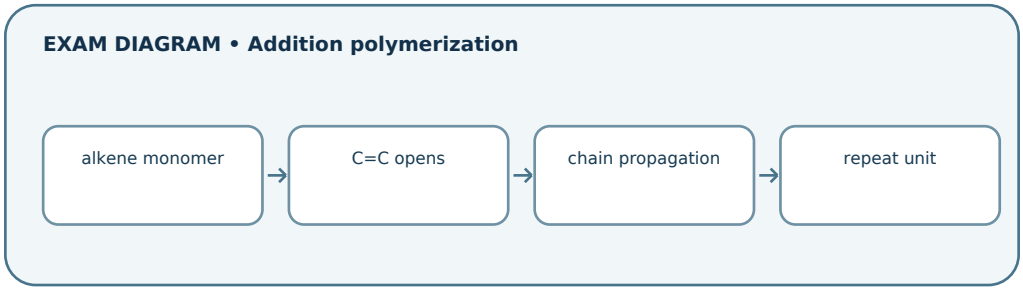
Answer: A. The double bond opens and each ethene molecule links to neighbouring units in the growing chain.

Question 2. In the diagram, which labelled step comes immediately after “alkene monomer”?

A. C=C opens
B. chain propagation
C. repeat unit
D. alkene monomer

Answer: A. The schematic is arranged in sequence: alkene monomer → C=C opens → chain propagation → repeat unit.

Exam reminder: Learn the chemical meaning of the arrows and labels; do not memorize a flowchart without understanding the reaction or process.



4.06 Stages of addition polymerization

Core concept

The textbook describes addition polymerization as a chain reaction. A growing polymer chain adds monomer units successively. Learn the general stages commonly used to explain chain-growth: initiation, propagation and termination. A catalyst or initiator may be involved depending on the system. Use the textbook's specific example and conditions when answering a detailed question.

Remember

Initiation → propagation → termination

Stages of addition polymerizatio	KEY RELATIONSHIP	EXAM APPLICATION

Exam focus

Be ready to define the term precisely, interpret the chemical representation, and explain the reason behind the result or process. Where a calculation is involved, identify the known quantities, choose the correct relationship, substitute with units and check whether the answer is chemically reasonable.

Alternative practice questions

A. Name the stages of chain-growth polymerization.

B. Describe what happens during propagation.

4.06 • Exam diagram & MCQ practice

Key exam idea: Chain-growth addition polymerization commonly involves initiation, propagation and termination.

Diagram reading: Redraw the sequence and explain the role of each labelled part. This is an original revision schematic, not a scale drawing.

Question 1. Which stage involves repeated addition of monomer to the growing chain?

A. propagation
B. initiation
C. termination
D. hydrolysis

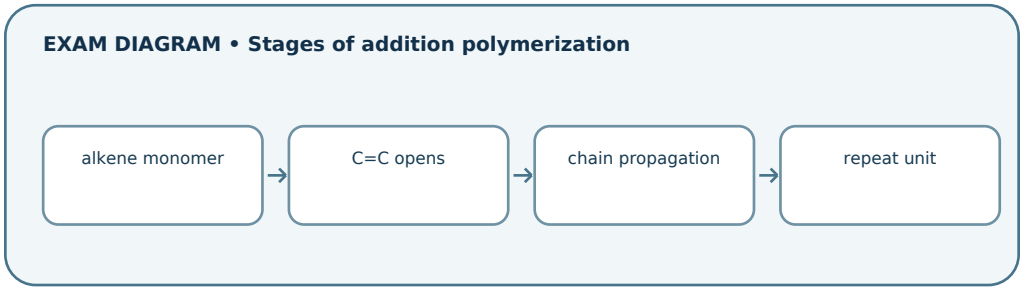
Answer: A. During propagation, active chain ends add further monomer units.

Question 2. In the diagram, which labelled step comes immediately after “alkene monomer”?

A. C=C opens
B. chain propagation
C. repeat unit
D. alkene monomer

Answer: A. The schematic is arranged in sequence: alkene monomer → C=C opens → chain propagation → repeat unit.

Exam reminder: Learn the chemical meaning of the arrows and labels; do not memorize a flowchart without understanding the reaction or process.



4.07 Polyethylene

Core concept

Polyethylene is formed by addition polymerization of ethene. It is lightweight and widely used for packaging, squeeze bottles, bags, tubing and electrical insulation. Its repeating unit is formed when the ethene double bond opens and links carbon atoms into a chain. Distinguish the monomer from the polymer structure.

Remember

Monomer: CH■=CH■; repeat unit: [–CH■–CH■–]■

Polyethylene	KEY RELATIONSHIP	EXAM APPLICATION

Exam focus

Be ready to define the term precisely, interpret the chemical representation, and explain the reason behind the result or process. Where a calculation is involved, identify the known quantities, choose the correct relationship, substitute with units and check whether the answer is chemically reasonable.

Alternative practice questions

A. Write the equation for formation of polyethylene.

B. Give three uses of polyethylene and relate one use to a property.

4.07 • Exam diagram & MCQ practice

Key exam idea: Polyethylene is formed by addition polymerization of ethene and is widely used in packaging and containers.

Diagram reading: Redraw the sequence and explain the role of each labelled part. This is an original revision schematic, not a scale drawing.

Question 1. Which repeat unit represents polyethylene?

A. –CH_2–CH_2–
B. –CH_2–CHCl–
C. –CF_2–CF_2–
D. –CH_2–$CH(CH_3)$–

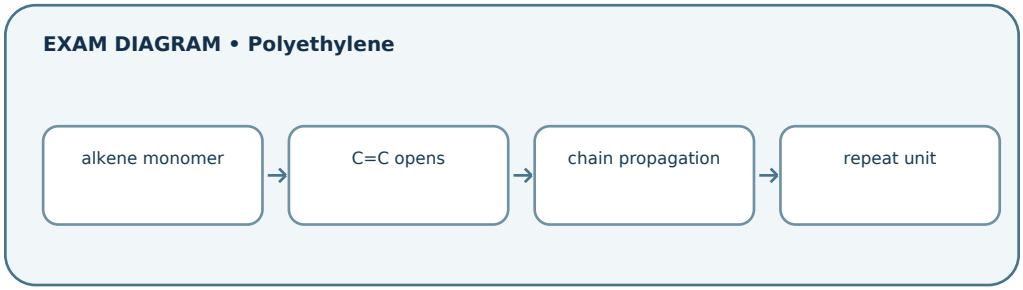
Answer: A. Ethene polymerizes to the repeating –CH_2–CH_2– unit.

Question 2. In the diagram, which labelled step comes immediately after “alkene monomer”?

A. C=C opens
B. chain propagation
C. repeat unit
D. alkene monomer

Answer: A. The schematic is arranged in sequence: alkene monomer → C=C opens → chain propagation → repeat unit.

Exam reminder: Learn the chemical meaning of the arrows and labels; do not memorize a flowchart without understanding the reaction or process.



4.08 Polystyrene

Core concept

Polystyrene is an addition polymer formed from styrene. The double bond in the monomer is used to build the polymer backbone while the phenyl group remains attached to the chain. The textbook includes it among common addition polymers. Be able to identify the monomer and the repeated structural feature.

Remember

Monomer: styrene (C■H■CH=CH■)

Polystyrene	KEY RELATIONSHIP	EXAM APPLICATION

Exam focus

Be ready to define the term precisely, interpret the chemical representation, and explain the reason behind the result or process. Where a calculation is involved, identify the known quantities, choose the correct relationship, substitute with units and check whether the answer is chemically reasonable.

Alternative practice questions

A. Identify the monomer of polystyrene.

B. Explain why styrene can undergo addition polymerization.

4.08 • Exam diagram & MCQ practice

Key exam idea: Polystyrene is an addition polymer of styrene; its repeat unit carries a phenyl side group.

Diagram reading: Redraw the sequence and explain the role of each labelled part. This is an original revision schematic, not a scale drawing.

Question 1. Which monomer forms polystyrene?

A. styrene
B. ethene
C. propene
D. vinyl chloride

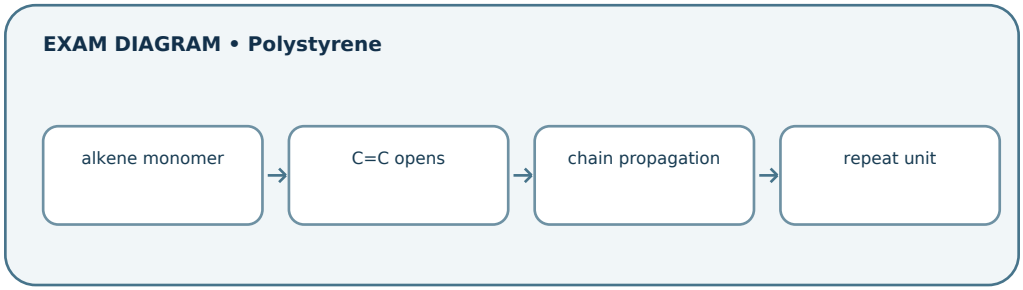
Answer: A. Styrene (phenylethene) undergoes addition polymerization through its carbon–carbon double bond.

Question 2. In the diagram, which labelled step comes immediately after “alkene monomer”?

A. C=C opens
B. chain propagation
C. repeat unit
D. alkene monomer

Answer: A. The schematic is arranged in sequence: alkene monomer → C=C opens → chain propagation → repeat unit.

Exam reminder: Learn the chemical meaning of the arrows and labels; do not memorize a flowchart without understanding the reaction or process.



4.09 Polypropylene

Core concept

Polypropylene is produced by addition polymerization of propene. Its repeat unit contains a methyl side group. The textbook describes it as stronger than polyethylene and notes uses including food containers, ropes, fishing nets, carpets and bottles. Link structure, properties and uses in exam responses.

Remember

Monomer: CH■=CHCH■; repeat unit: [–CH■–CH(CH■)–]■

Polypropylene	KEY RELATIONSHIP	EXAM APPLICATION

Exam focus

Be ready to define the term precisely, interpret the chemical representation, and explain the reason behind the result or process. Where a calculation is involved, identify the known quantities, choose the correct relationship, substitute with units and check whether the answer is chemically reasonable.

Alternative practice questions

A. Write the repeat unit of polypropylene.

B. State two uses of polypropylene and one relevant property.

4.09 • Exam diagram & MCQ practice

Key exam idea: Polypropylene is made by addition polymerization of propene and has a methyl side group on alternate backbone carbons.

Diagram reading: Redraw the sequence and explain the role of each labelled part. This is an original revision schematic, not a scale drawing.

Question 1. Which monomer produces polypropylene?

A. propene
B. ethene
C. chloroethene
D. tetrafluoroethene

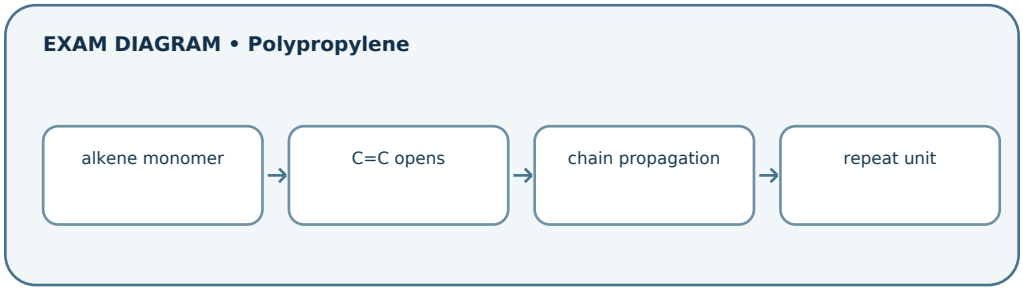
Answer: A. Propene polymerizes to a chain with –CH_2–$CH(CH_3)$– repeat units.

Question 2. In the diagram, which labelled step comes immediately after “alkene monomer”?

A. C=C opens
B. chain propagation
C. repeat unit
D. alkene monomer

Answer: A. The schematic is arranged in sequence: alkene monomer → C=C opens → chain propagation → repeat unit.

Exam reminder: Learn the chemical meaning of the arrows and labels; do not memorize a flowchart without understanding the reaction or process.



4.10 Polyvinyl chloride (PVC)

Core concept

PVC is formed by polymerizing vinyl chloride (chloroethene). Chlorine atoms are attached to the carbon backbone. The textbook lists pipes, leather-like materials, shoes, raincoats, aprons, wallpaper and floor tiles among its uses. Recognize the monomer and identify the chlorine substituent in the repeat unit.

Remember

Monomer: CH■=CHCl; repeat unit: [–CH■–CHCl–]■

Polyvinyl chloride (PVC)	KEY RELATIONSHIP	EXAM APPLICATION

Exam focus

Be ready to define the term precisely, interpret the chemical representation, and explain the reason behind the result or process. Where a calculation is involved, identify the known quantities, choose the correct relationship, substitute with units and check whether the answer is chemically reasonable.

Alternative practice questions

A. Name the monomer used to produce PVC.

B. Give three uses of PVC from the textbook.

4.10 • Exam diagram & MCQ practice

Key exam idea: PVC is formed by addition polymerization of chloroethene (vinyl chloride); chlorine atoms are attached to the backbone.

Diagram reading: Redraw the sequence and explain the role of each labelled part. This is an original revision schematic, not a scale drawing.

Question 1. The monomer of PVC is:

A. chloroethene
B. ethene
C. styrene
D. propene

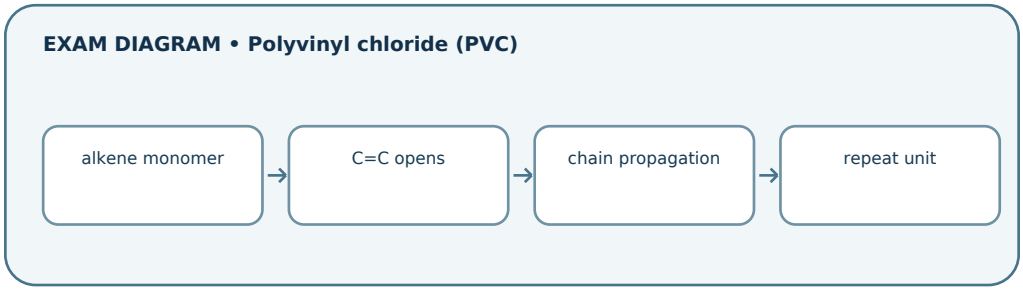
Answer: A. Chloroethene, $CH_2=CHCl$, forms the $–CH_2–CHCl–$ repeat unit.

Question 2. In the diagram, which labelled step comes immediately after “alkene monomer”?

A. C=C opens
B. chain propagation
C. repeat unit
D. alkene monomer

Answer: A. The schematic is arranged in sequence: alkene monomer → C=C opens → chain propagation → repeat unit.

Exam reminder: Learn the chemical meaning of the arrows and labels; do not memorize a flowchart without understanding the reaction or process.



4.11 PMMA and PTFE

Core concept

Polymethyl methacrylate (PMMA) is a lightweight, glass-like polymer used as a glass substitute, including in aircraft windows and streetlights. PTFE (Teflon) is made from tetrafluoroethene and resists chemical attack; it has a low coefficient of friction and is used in non-stick coatings and electrical insulation. Link each material to its monomer and properties.

Remember

PMMA: methyl methacrylate; PTFE: CF■=CF■

PMMA and PTFE	KEY RELATIONSHIP	EXAM APPLICATION

Exam focus

Be ready to define the term precisely, interpret the chemical representation, and explain the reason behind the result or process. Where a calculation is involved, identify the known quantities, choose the correct relationship, substitute with units and check whether the answer is chemically reasonable.

Alternative practice questions

A. Which polymer is used as a glass substitute, and why?

B. Name the monomer and two properties of PTFE.

4.11 • Exam diagram & MCQ practice

Key exam idea: PMMA is a transparent acrylic glass substitute; PTFE is chemically resistant and has a low-friction surface.

Diagram reading: Redraw the sequence and explain the role of each labelled part. This is an original revision schematic, not a scale drawing.

Question 1. Which polymer is commonly used as a transparent glass substitute?

A. PMMA
B. PTFE
C. Bakelite
D. nylon-6,6

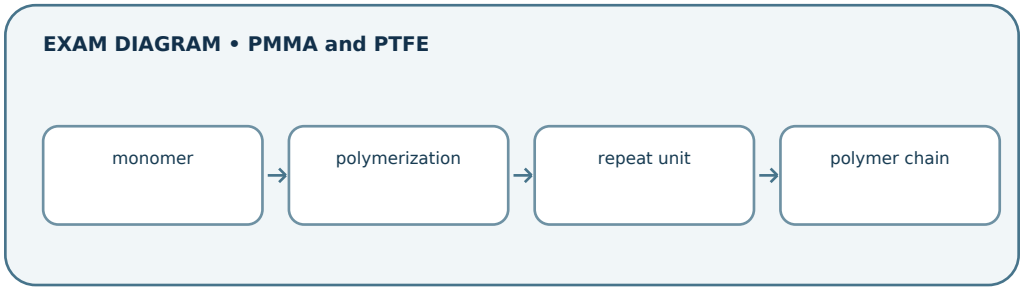
Answer: A. PMMA (acrylic) is transparent and lightweight, so it can replace glass in some applications.

Question 2. In the diagram, which labelled step comes immediately after “monomer”?

A. polymerization
B. repeat unit
C. polymer chain
D. monomer

Answer: A. The schematic is arranged in sequence: monomer → polymerization → repeat unit → polymer chain.

Exam reminder: Learn the chemical meaning of the arrows and labels; do not memorize a flowchart without understanding the reaction or process.



4.12 Condensation polymerization

Core concept

Condensation polymerization joins monomers through reactions between functional groups and commonly eliminates a small molecule such as water. It is generally associated with step-growth formation. The textbook examples include nylon, polyester and Bakelite. Identify the functional groups and by-product when a particular condensation reaction is shown.

Remember

Monomers + functional groups → polymer + small molecule

Condensation polymerization	KEY RELATIONSHIP	EXAM APPLICATION

Exam focus

Be ready to define the term precisely, interpret the chemical representation, and explain the reason behind the result or process. Where a calculation is involved, identify the known quantities, choose the correct relationship, substitute with units and check whether the answer is chemically reasonable.

Alternative practice questions

A. Define condensation polymerization.

B. How can a reaction equation reveal that polymerization is condensation rather than addition?

4.12 • Exam diagram & MCQ practice

Key exam idea: Condensation polymerization joins functional monomers while eliminating a small molecule such as water or HCl.

Diagram reading: Redraw the sequence and explain the role of each labelled part. This is an original revision schematic, not a scale drawing.

Question 1. Which observation is characteristic of condensation polymerization?

A. a small molecule is eliminated during linkage formation
B. only C=C bonds open and no by-product forms
C. monomers remain unconnected
D. the polymer always contains fluorine

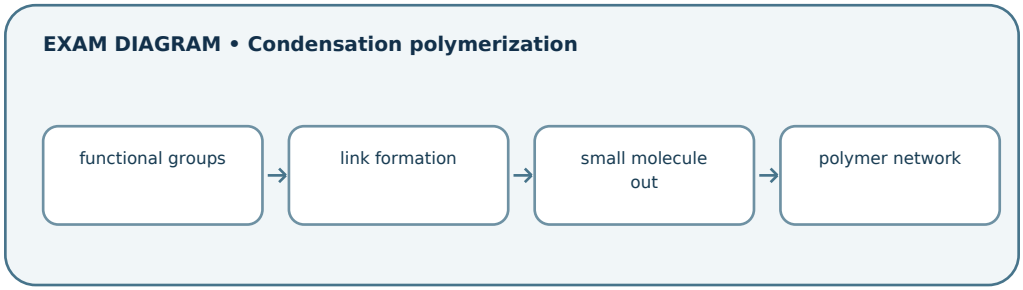
Answer: A. Condensation reactions form links between functional groups and commonly release a small molecule.

Question 2. In the diagram, which labelled step comes immediately after “functional groups”?

A. link formation
B. small molecule out
C. polymer network
D. functional groups

Answer: A. The schematic is arranged in sequence: functional groups → link formation → small molecule out → polymer network.

Exam reminder: Learn the chemical meaning of the arrows and labels; do not memorize a flowchart without understanding the reaction or process.



4.13 Nylon

Core concept

Nylon is a synthetic polyamide made by condensation polymerization. Amide linkages connect the repeating units. Its properties make it useful as a synthetic fibre and engineering material. When asked about nylon, identify its polymer class, the linkage between units and practical uses. Avoid confusing it with polyester, which contains ester linkages.

Remember

Characteristic linkage: –CONH–

Nylon	KEY RELATIONSHIP	EXAM APPLICATION

Exam focus

Be ready to define the term precisely, interpret the chemical representation, and explain the reason behind the result or process. Where a calculation is involved, identify the known quantities, choose the correct relationship, substitute with units and check whether the answer is chemically reasonable.

Alternative practice questions

A. What type of polymer is nylon?

B. Identify the characteristic linkage in a polyamide.

4.13 • Exam diagram & MCQ practice

Key exam idea: Nylon is a synthetic polyamide; amide linkages connect its repeating units.

Diagram reading: Redraw the sequence and explain the role of each labelled part. This is an original revision schematic, not a scale drawing.

Question 1. Which linkage is characteristic of nylon?

A. amide
B. ester
C. ether only
D. ionic

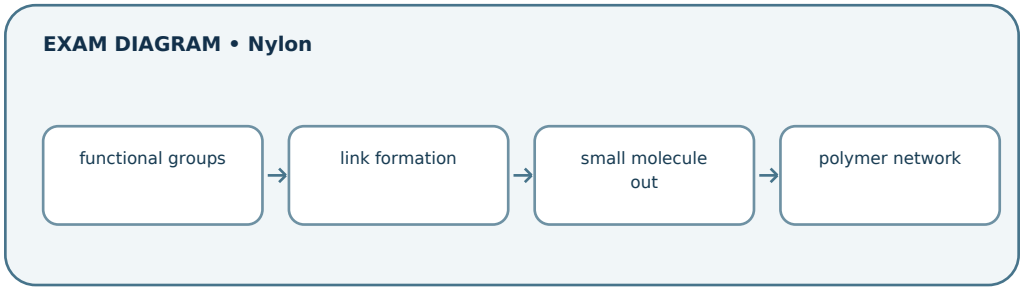
Answer: A. Nylon contains repeating amide links, –CO–NH–, in its polymer backbone.

Question 2. In the diagram, which labelled step comes immediately after “functional groups”?

A. link formation
B. small molecule out
C. polymer network
D. functional groups

Answer: A. The schematic is arranged in sequence: functional groups → link formation → small molecule out → polymer network.

Exam reminder: Learn the chemical meaning of the arrows and labels; do not memorize a flowchart without understanding the reaction or process.



4.14 Polyester

Core concept

Polyesters contain ester linkages in their polymer chains and are formed through condensation reactions involving suitable functional groups. The textbook discusses polyester among condensation polymers and gives Dacron as an example. Uses include fibres and textile materials. Compare the ester linkage with the amide linkage in nylon.

Remember

Characteristic linkage: –COO–

Polyester	KEY RELATIONSHIP	EXAM APPLICATION

Exam focus

Be ready to define the term precisely, interpret the chemical representation, and explain the reason behind the result or process. Where a calculation is involved, identify the known quantities, choose the correct relationship, substitute with units and check whether the answer is chemically reasonable.

Alternative practice questions

A. What linkage identifies a polyester?

B. Compare polyester and nylon by their characteristic linkages.

4.14 • Exam diagram & MCQ practice

Key exam idea: Polyesters contain ester linkages and are commonly made by condensation reactions of suitable monomers.

Diagram reading: Redraw the sequence and explain the role of each labelled part. This is an original revision schematic, not a scale drawing.

Question 1. Which functional linkage identifies a polyester?

A. –COO–
B. –CONH–
C. –C=C–
D. –O–O–

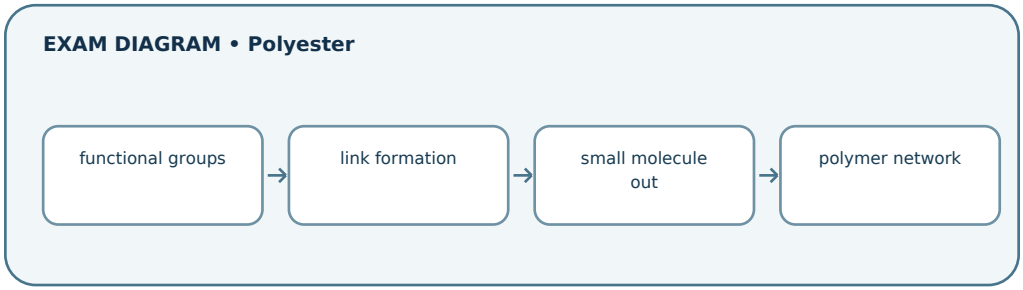
Answer: A. The ester group –COO– joins the repeating units in polyester chains.

Question 2. In the diagram, which labelled step comes immediately after “functional groups”?

A. link formation
B. small molecule out
C. polymer network
D. functional groups

Answer: A. The schematic is arranged in sequence: functional groups → link formation → small molecule out → polymer network.

Exam reminder: Learn the chemical meaning of the arrows and labels; do not memorize a flowchart without understanding the reaction or process.



4.15 Bakelite

Core concept

Bakelite is a condensation polymer and a thermosetting material. Its cross-linked network structure makes it hard and resistant to softening on reheating. The textbook includes Bakelite with nylon and polyester as an important condensation polymer example. Relate its network structure to its thermal behavior and uses.

Remember

Cross-linked thermoset network

Bakelite	KEY RELATIONSHIP	EXAM APPLICATION

Exam focus

Be ready to define the term precisely, interpret the chemical representation, and explain the reason behind the result or process. Where a calculation is involved, identify the known quantities, choose the correct relationship, substitute with units and check whether the answer is chemically reasonable.

Alternative practice questions

A. Why is Bakelite classified as a thermosetting polymer?

B. Explain how cross-linking affects the behavior of Bakelite when heated.

4.15 • Exam diagram & MCQ practice

Key exam idea: Bakelite is a highly cross-linked thermosetting polymer that becomes rigid and does not remelt on reheating.

Diagram reading: Redraw the sequence and explain the role of each labelled part. This is an original revision schematic, not a scale drawing.

Question 1. Why does Bakelite not soften and flow like a thermoplastic when reheated?

A. its chains form a rigid cross-linked network
B. it contains no covalent bonds
C. it consists only of monomers
D. it dissolves in water

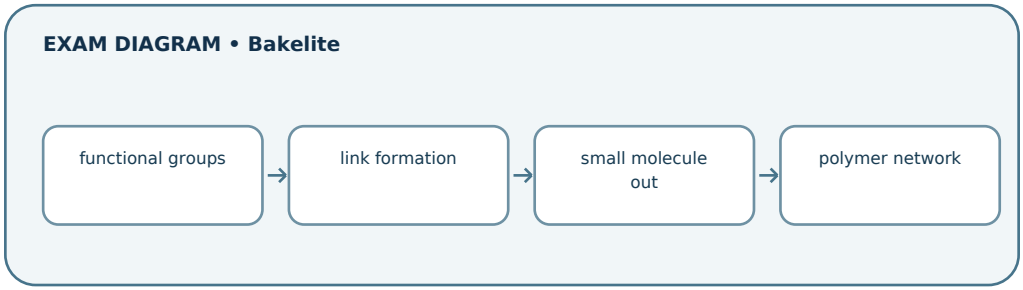
Answer: A. Extensive cross-links prevent polymer chains from sliding past one another when heated.

Question 2. In the diagram, which labelled step comes immediately after “functional groups”?

A. link formation
B. small molecule out
C. polymer network
D. functional groups

Answer: A. The schematic is arranged in sequence: functional groups → link formation → small molecule out → polymer network.

Exam reminder: Learn the chemical meaning of the arrows and labels; do not memorize a flowchart without understanding the reaction or process.



4.16 Natural polymers

Core concept

Natural polymers occur in living organisms and natural materials. Textbook examples include cellulose in wood, proteins and DNA. They are built from biological monomers such as glucose and amino acids. Identify the repeating building block and the biological role or source where stated. Do not assume all natural polymers have the same structure or properties.

Remember

Cellulose: glucose units; proteins: amino acids

Natural polymers	KEY RELATIONSHIP	EXAM APPLICATION

Exam focus

Be ready to define the term precisely, interpret the chemical representation, and explain the reason behind the result or process. Where a calculation is involved, identify the known quantities, choose the correct relationship, substitute with units and check whether the answer is chemically reasonable.

Alternative practice questions

A. Give three examples of natural polymers.

B. Identify the monomer associated with cellulose and the building blocks of proteins.

4.16 • Exam diagram & MCQ practice

Key exam idea: Natural polymers occur in living organisms; examples include cellulose, starch, proteins and natural rubber.

Diagram reading: Redraw the sequence and explain the role of each labelled part. This is an original revision schematic, not a scale drawing.

Question 1. Which is a natural polymer?

A. cellulose
B. polyvinyl chloride
C. polystyrene
D. PTFE

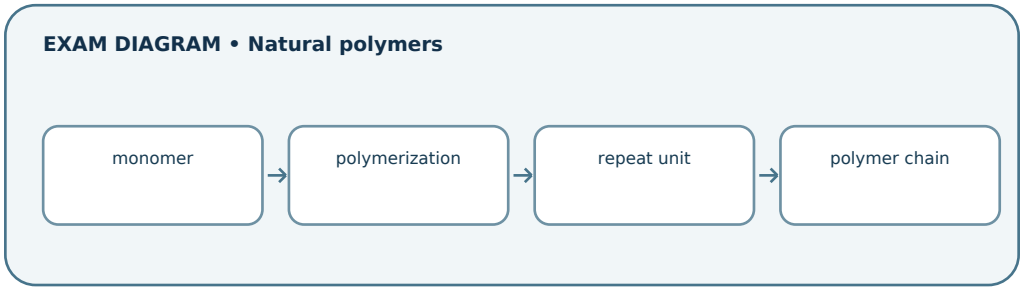
Answer: A. Cellulose is a naturally occurring polysaccharide found in plant cell walls.

Question 2. In the diagram, which labelled step comes immediately after “monomer”?

A. polymerization
B. repeat unit
C. polymer chain
D. monomer

Answer: A. The schematic is arranged in sequence: monomer → polymerization → repeat unit → polymer chain.

Exam reminder: Learn the chemical meaning of the arrows and labels; do not memorize a flowchart without understanding the reaction or process.



4.17 Synthetic polymers

Core concept

Synthetic polymers are human-made materials, many of which are commonly called plastics. The textbook discusses polyethylene, polypropylene, PVC, PMMA and PTFE. Their properties and uses vary with the monomer and polymer structure. For exam comparisons, organize examples by monomer, polymer class, properties and applications.

Remember

Human-made polymer examples: PE, PP, PVC, PMMA, PTFE

Synthetic polymers	KEY RELATIONSHIP	EXAM APPLICATION

Exam focus

Be ready to define the term precisely, interpret the chemical representation, and explain the reason behind the result or process. Where a calculation is involved, identify the known quantities, choose the correct relationship, substitute with units and check whether the answer is chemically reasonable.

Alternative practice questions

A. Distinguish natural and synthetic polymers with examples.

B. Match polyethylene, PVC and PTFE with one use each.

4.17 • Exam diagram & MCQ practice

Key exam idea: Synthetic polymers are human-made; examples include polyethylene, PVC, polystyrene and nylon.

Diagram reading: Redraw the sequence and explain the role of each labelled part. This is an original revision schematic, not a scale drawing.

Question 1. Which group contains only synthetic polymers?

A. polyethylene, PVC and nylon
B. cellulose, starch and proteins
C. DNA, cellulose and natural rubber
D. starch, wool and silk

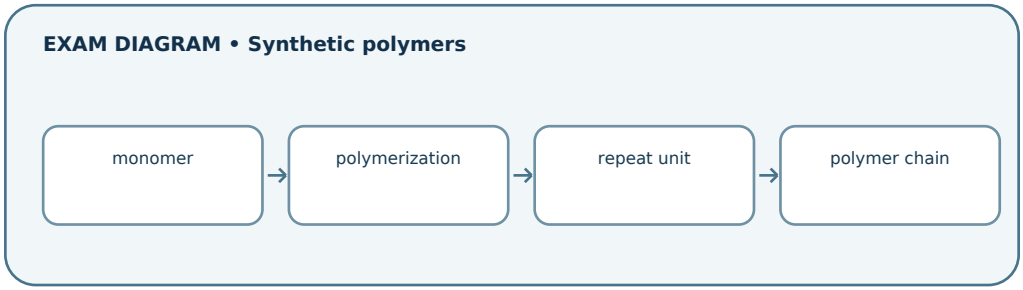
Answer: A. Polyethylene, PVC and nylon are manufactured polymers; the other choices include natural polymers.

Question 2. In the diagram, which labelled step comes immediately after “monomer”?

A. polymerization
B. repeat unit
C. polymer chain
D. monomer

Answer: A. The schematic is arranged in sequence: monomer → polymerization → repeat unit → polymer chain.

Exam reminder: Learn the chemical meaning of the arrows and labels; do not memorize a flowchart without understanding the reaction or process.



4.18 Thermoplastics

Core concept

Thermoplastics soften or melt when heated and solidify when cooled. Their molecules are mainly linear or lightly branched, allowing them to be remelted and reprocessed. Examples listed in the textbook include polyethylene, polypropylene, PVC, PTFE, PMMA, nylon and polyester. Their ability to be reprocessed can support recycling.

Remember

Heat → soften; cool → solidify

Thermoplastics	KEY RELATIONSHIP	EXAM APPLICATION

Exam focus

Be ready to define the term precisely, interpret the chemical representation, and explain the reason behind the result or process. Where a calculation is involved, identify the known quantities, choose the correct relationship, substitute with units and check whether the answer is chemically reasonable.

Alternative practice questions

A. Define thermoplastic polymer and give three examples.

B. Why can many thermoplastics be remoulded?

4.18 • Exam diagram & MCQ practice

Key exam idea: Thermoplastics soften on heating and harden on cooling, so many can be remoulded repeatedly.

Diagram reading: Redraw the sequence and explain the role of each labelled part. This is an original revision schematic, not a scale drawing.

Question 1. Which property is typical of a thermoplastic?

A. it can soften on heating and be remoulded
B. it forms an irreversible rigid network
C. it cannot be shaped when manufactured
D. it decomposes immediately in cold water

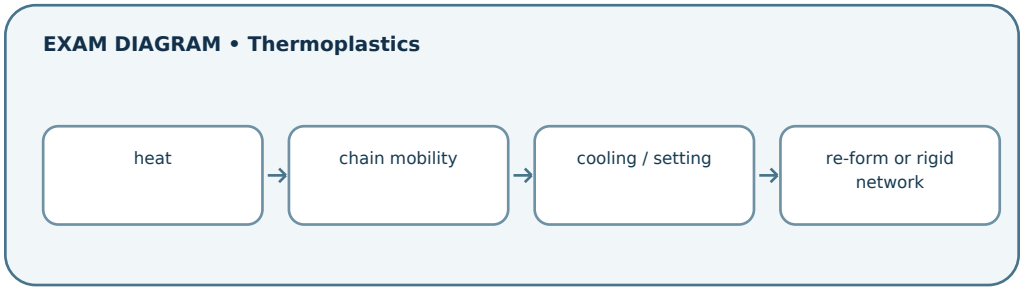
Answer: A. Thermoplastic chains can move past one another when heated, allowing reshaping.

Question 2. In the diagram, which labelled step comes immediately after “heat”?

A. chain mobility
B. cooling / setting
C. re-form or rigid network
D. heat

Answer: A. The schematic is arranged in sequence: heat → chain mobility → cooling / setting → re-form or rigid network.

Exam reminder: Learn the chemical meaning of the arrows and labels; do not memorize a flowchart without understanding the reaction or process.



4.19 Thermosetting polymers

Core concept

Thermosetting polymers form rigid, cross-linked networks. Once set, they do not simply melt and flow again on heating; excessive heating may cause decomposition. Bakelite is a textbook example. Compare thermosets with thermoplastics by molecular structure and response to heat, not merely by whether they are hard.

Remember

Cross-linked network; cannot be remelted into a new shape

Thermosetting polymers	KEY RELATIONSHIP	EXAM APPLICATION

Exam focus

Be ready to define the term precisely, interpret the chemical representation, and explain the reason behind the result or process. Where a calculation is involved, identify the known quantities, choose the correct relationship, substitute with units and check whether the answer is chemically reasonable.

Alternative practice questions

A. Compare thermoplastics and thermosets.

B. Explain how cross-linking accounts for thermosetting behavior.

4.19 • Exam diagram & MCQ practice

Key exam idea: Thermosetting polymers form strongly cross-linked networks and cannot be remelted and remoulded after setting.

Diagram reading: Redraw the sequence and explain the role of each labelled part. This is an original revision schematic, not a scale drawing.

Question 1. Which description best fits a thermosetting polymer?

A. rigid, cross-linked network
B. linear chains that repeatedly melt
C. small molecules with no covalent bonds
D. a polymer that is always water-soluble

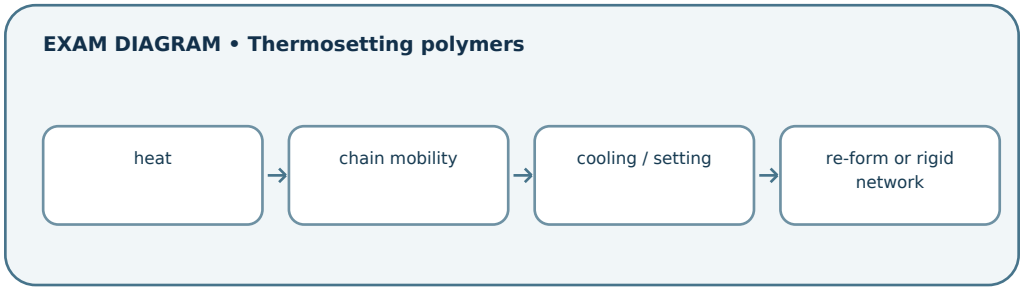
Answer: A. Extensive cross-linking locks the chains into a network, so the material does not flow on reheating.

Question 2. In the diagram, which labelled step comes immediately after “heat”?

A. chain mobility
B. cooling / setting
C. re-form or rigid network
D. heat

Answer: A. The schematic is arranged in sequence: heat → chain mobility → cooling / setting → re-form or rigid network.

Exam reminder: Learn the chemical meaning of the arrows and labels; do not memorize a flowchart without understanding the reaction or process.



4.20 Polymer identification and exam synthesis

Core concept

For an unfamiliar polymer question, identify the monomer, locate unsaturation or functional groups, decide addition versus condensation, draw the repeat unit and classify the material. Then connect structure to properties and uses. The strongest answers show the chemical reasoning rather than memorizing product names alone.

Remember

Monomer → reaction type → repeat unit → class → properties/use

Polymer identification and exam	KEY RELATIONSHIP	EXAM APPLICATION

Exam focus

Be ready to define the term precisely, interpret the chemical representation, and explain the reason behind the result or process. Where a calculation is involved, identify the known quantities, choose the correct relationship, substitute with units and check whether the answer is chemically reasonable.

Alternative practice questions

A. Given an alkene monomer, predict the polymerization type and repeat unit.

B. A polymer contains ester links and is made with elimination of a small molecule. Identify its polymerization type.

4.20 • Exam diagram & MCQ practice

Key exam idea: Identify the monomer and functional groups, then decide whether the polymer forms by addition or condensation.

Diagram reading: Redraw the sequence and explain the role of each labelled part. This is an original revision schematic, not a scale drawing.

Question 1. A polymer contains ester links and forms with elimination of a small molecule. It is most likely made by:

A. condensation polymerization
B. addition polymerization
C. simple crystallization
D. electrolysis

Answer: A. Ester link formation with elimination of a small molecule is characteristic of condensation polymerization.

Question 2. In the diagram, which labelled step comes immediately after “monomer”?

A. polymerization
B. repeat unit
C. polymer chain
D. monomer

Answer: A. The schematic is arranged in sequence: monomer → polymerization → repeat unit → polymer chain.

Exam reminder: Learn the chemical meaning of the arrows and labels; do not memorize a flowchart without understanding the reaction or process.

